Ryan Hartung

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Education

The University of Notre Dame — Notre Dame, IN

Ph.D. in Computer Science & Engineering — *Expected May 2029*

- Advisor: Dr. Douglas Thain
- Research Focus: Portability, Reliability, and Scalability of Distributed and High-Performance Systems

M.S. in Computer Science & Engineering — *Aug 2024 – Aug 2025*

- Master's Project: HPC integration and multi-site portability for the xGFabric framework

The University of Texas at Austin — Austin, TX

B.S. in Physics (Computation Option) — *Aug 2020 – May 2024*

- Certificate in Elements of Computing

Publications

[2026] Liubov Kurafeeva, **Ryan Hartung**, Benjamin Carter, Alan Subedi, Avhishek Biswas, Michael Fay, Shantenu Jha, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "Asynchronous Scientific Workflows for Simulation-Driven Inference Across the Edge–HPC Computing Continuum." *Future Generation Computer Systems* (Elsevier), special issue, 2026. (Under review).

[2026] Liubov Kurafeeva, Benjamin Carter, **Ryan Hartung**, Alan Subedi, Avhishek Biswas, Michael Fay, Shantenu Jha, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "GROWFLOW: Low-latency Computational Fluid Dynamics at the Edge for Real-time Agricultural Decision Support." In *Proceedings of IEEE Smart Connected Systems and Emerging Technologies Conference (IEEE SCSETech)*, 2026 (To appear).

[2026] Liubov Kurafeeva, **Ryan Hartung**, Benjamin Carter, Alan Subedi, Avhishek Biswas, Michael Fay, Shantenu Jha, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference." *arXiv preprint arXiv:2605.20532*. https://arxiv.org/abs/2605.20532

[2025] Liubov Kurafeeva, Alan Subedi, **Ryan Hartung**, Michael Fay, Avhishek Biswas, Shantenu Jha, Ozgur Kilic, Chandra Krintz, Andre Merzky, Douglas Thain, Mehmet Vuran, and Rich Wolski. "xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture." In *Proceedings of the SC '25 Workshops*. ACM, 2317–2327. DOI: https://doi.org/10.1145/3731599.3767589

Research Experience

The University of Notre Dame — Notre Dame, IN

Graduate Research Assistant, Cooperative Computing Lab — *Jan 2025 – Present*

- Engineered the HPC integration pipeline for xGFabric, enabling real-time streaming of remote 5G sensor data directly into Computational Fluid Dynamics (CFD) simulations.
- Standardized multi-site deployment across 4 supercomputing facilities (ND CRC, Purdue Anvil, TACC Stampede3, NERSC Perlmutter), resolving cross-platform incompatibilities in hardware, OSs, and batch schedulers.
- Converted sequential Bash scripts into a production-ready Makeflow/Work Queue workflow, co-scheduling 72 concurrent OpenFOAM jobs at NERSC and meeting the 2-hour turnaround deadline for surrogate ML model inference (PINN/FNO/PCR).
- Ported the workflow to HTCondor and implemented a "wait-for-workers" feature in Work Queue and TaskVine (CCTools), bringing `salloc`-style co-scheduling to throughput computing.
- Built a proof-of-concept Random Forest model predicting queue wait times for 4–64 core jobs, trained on multi-month queue data collected across 3 HPC sites.
- Identified OpenFOAM simulation as the pipeline's latency bottleneck; profiled scaling from 1–64 cores across 1–4 nodes and found single-node 32–64 core runs minimized turnaround, outperforming multi-node runs.

The University of Texas at Austin — Austin, TX

Undergraduate Research Assistant — *June 2022 – December 2023*

- Engineered an open-source Python tool analyzing star-spot patterns across 6,000+ star systems with optimized array processing algorithms.
- Evaluated star-spot contrast ratios using observational data from NASA Kepler and TESS missions to establish baseline metrics for exoplanet detection.
- Integrated parallel processing and multi-core compilation to optimize data throughput and execution runtime.
- Published codebase and workflows to an open-source GitHub repository; research funded by NASA grant.

Work Experience

Hargrove Engineers and Constructors Katy, TX

Controls & Automations Engineering Intern *May 2024 – August 2024*

- Automated the migration of 200,000+ lines of Dow MOD5 code into Honeywell Experion with no loss of functionality.
- Tracked, analyzed, and documented over 7,000 variables to design flow diagrams visualizing variable definitions.
- Developed several algorithms to expedite the migration and variable mapping processes, improving overall efficiency.

Hargrove Engineers and Constructors Katy, TX

Controls & Automations Engineering Intern *May 2023 – August 2023*

- Instantiated existing library modules to complete Westlake's PVC plant Reactor and Train configuration.
- Performed safety testing on all module and sequence configurations on simulated ABB hardware.
- Successfully configured Modbus communication between the ABB and HIMA system controllers.
- Updated structured text configuration to handle cross-controller communications.

Harris County Houston, TX

Elections Clerk *May 2022 & Dec 2023*

- Assisted with polling operations, equipment setup and takedown, and ballot transportion.

Talks, Panels, & Presentations

GCASR 2026 Evanston, IL

Poster Presenter *May 11, 2026*

- Poster: *Hybrid Edge-HPC Systems for Low-Latency Data-Driven Inference*.

Supercomputing '25 St. Louis, MO

Presenter, Demonstrator, and Panelist *November 16 – 19, 2025*

- Featured panelist for the XLOOP Workshop expert session on edge-to-HPC loop architectures and computing trends [Recording available online].
- Presented xGFabric and its findings at the Department of Energy's (DOE) booth using a live demo to an audience of DOE staff, SC attendees, and fellow researchers. The demo showcased the full end-to-end pipeline: fetching and parsing data, launching simulations, and rendering results.

HPDC 2025 Notre Dame, IN

Poster Presenter *July 21, 2025*

- Poster: *xGFabric: Coupling Sensor Networks and HPC Facilities with 5G Wireless Networks for Real-Time Digital Agriculture*.

Teaching Experience

The University of Notre Dame Notre Dame, IN

Graduate Teaching Assistant *August 2024 – May 2025*

- CSE 10102; Elements of Computing II (Spring 2025): 37 students; held office hours, graded, assisted with in-class coding.
- CSE 10001; Principles of Computing (Fall 2024): 35 students; proctored exams and graded.

Technical Skills

Programming Languages: Bash, C, CUDA, Java, Python, Rust, SQL
HPC & Distributed Systems: HTCondor, Makeflow, MPI, OpenMP, Slurm, Univa Grid Engine (UGE), Work Queue
Technologies: Docker, Git, Linux/UNIX, Make/CMake, OpenFOAM, ParaView, Spack, Vim
Web Development: CSS, HTML, JavaScript/NodeJS, Jekyll, React

Projects

High-Performance Distributed Systems | *C, CUDA, OpenMP, HTCondor* *Aug 2026 – Present*

- Wrote CUDA and hybrid CPU+GPU versions of three benchmarks; hybrid versions ran 419× (heat) and 77× (wave) faster than serial CPU, and up to 13× faster than GPU-only.
- Characterized CPU/GPU crossover: kernel launch and transfer overhead let the CPU outperform the GPU on grids below ∼128–256 elements.
- Measured compiler auto-vectorization on a 1D heat solver (`-O3` gave ∼7× over unoptimized code; SIMD ISA flags had negligible effect) and OpenMP N-body scaling across sockets.

Bminor Compiler | *C, Rust, x86 Assembly* *Oct 2025 – Present*

- Ported a multi-stage C compiler to idiomatic Rust, implementing lexing, AST parsing, static type checking, and x86-64 code generation.
- Refactored C's legacy pointer-heavy data structures into memory-safe pattern matching and algebraic types.

Linux Auto-scroller | *Rust* *Jun 2025 – Present*

- Engineered a low-level system utility bringing native middle-click auto-scrolling to Linux.
- Interfaced with system input APIs to calculate cursor displacement and dynamically modulate scroll speed.

Leadership & Service

Order of Christian Initiation of Adults (OCIA), Sponsor — September 2026 – Present
CCL Web & Communications: Writing blog posts, publishing lab announcements, and maintaining the Cooperative Computing Lab's website. — May 2026 – Present
Fischer Graduate Residence Mass, Lector — August 2025 – May 2026
Society of Catholic Scientists, Member — August 2025 – Present
Science Policy Initiative at Notre Dame, Member — August 2025 – Present
Catholic Graduate Community, Member — April 2025 – Present
Technology and Catholicism Club, Member — February 2025 – Present
Notre Dame Rock Climbing Club, Member — August 2024 – Present
Notre Dame Texas Club, Member — August 2024 – Present
Science Olympiad, Competitor, Mentor, President, Event Supervisor, and Lab Coordinator — August 2011 – Present